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A large, dark blue, stylized lowercase letter 'i' logo, composed of a solid circle for the dot and a thick vertical stroke for the body, positioned on the left side of the slide.

UTTER

NAVER LABS Europe's Multilingual Speech Translation Systems for the IWSLT 2023 Low-Resource Track

Presenter: Marcelly Zanon Boito

Research Scientist - NAVER LABS Europe

Work done by: Edward Gow-Smith, Alexandre Bérard, Marcelly Zanon Boito, Ioan Calapodescu

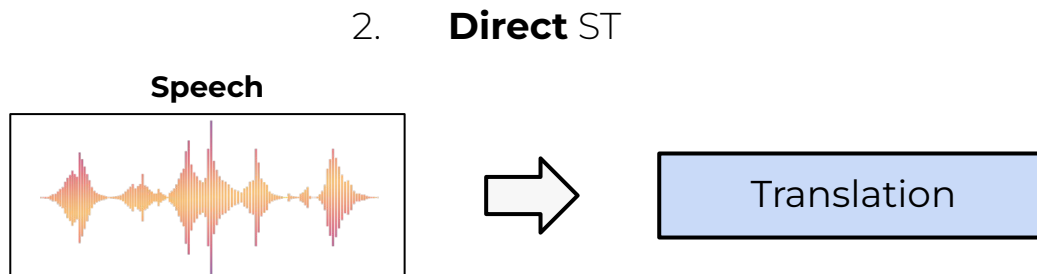
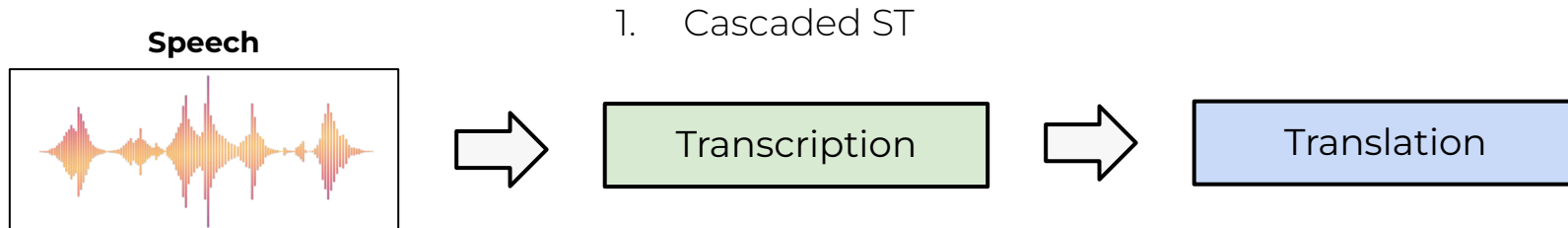
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Presentation Outline

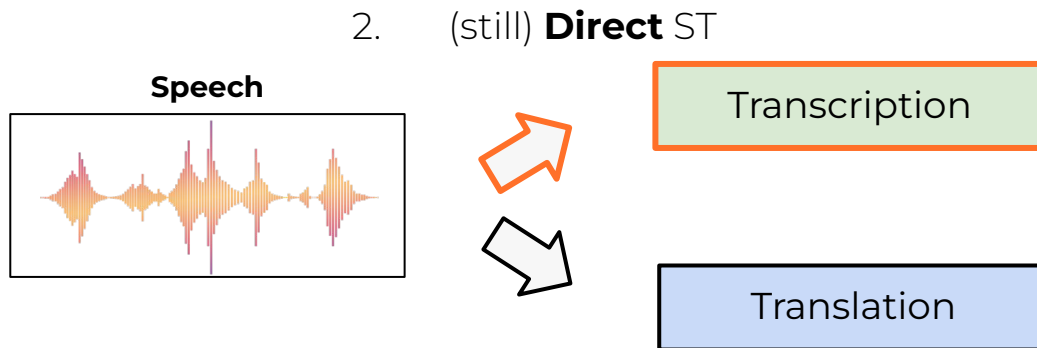
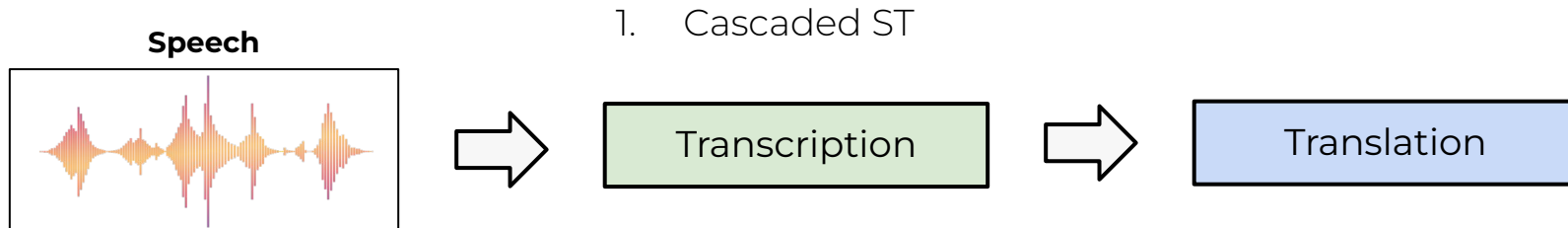
1. **Speech Translation for Low-resource Languages**
2. **NAVER LABS Europe IWSLT 2023 Systems**
3. **Results**

Speech Translation for Low-resource Languages

Speech Translation (ST):



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Direct Speech Translation:

- Why?
 - No error propagation: context is on the signal
 - Oral/Unwritten languages do not have standard transcript
 - Simultaneous settings require fast decoding

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IWSLT 2022: Low-resource and Dialect Track

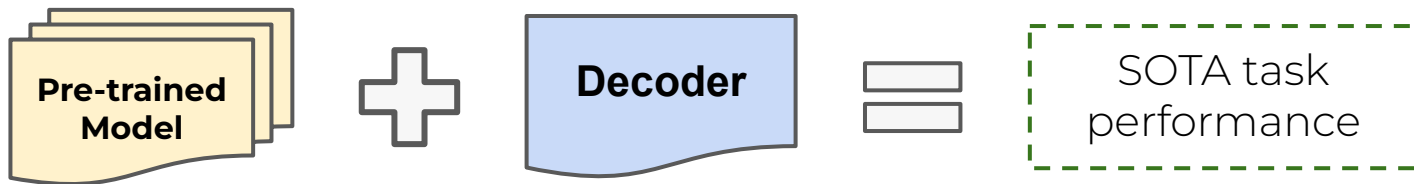
Team	System	Pre-trained Models	BLEU	chrF++
ON-TRAC	primary	wav2vec 2.0 (Tamasheq)	5.7	31.4
	contrastive	ASR (French)	5.0	26.7
TalTech	primary	XLS-R, mBART-50	2.7	24.3
GMU	primary	XLS-R (Arabic, French)	0.5	16.9

IWSLT 2022 introduced the **Tamasheq-French** language pair.

17h of speech with no transcripts, only translations.

The best submission [1] scored 5.7 BLEU. In *comparison**, submissions to offline track start at 25 BLEU.

Why such a gap?



- **Only 17h of Tamasheq.** With noise. And code-switching
- Dedicated pre-trained models for speech help, but they are not enough
- “Cold decoder” does not produce fluent French

NAVER LABS Europe IWSLT 2023 Systems

IWSLT 2023 Low-resource and Dialect Track

Target:

- Tamasheq-French (17h)
- Quechua-Spanish (2h) + (51h of ASR)

Goal:

1. One single system **(multilingual)**
2. Work on top of existing pre-trained models **(parameter efficient)**
3. Build something useful for speech and text **(multimodal)**

Our Architecture

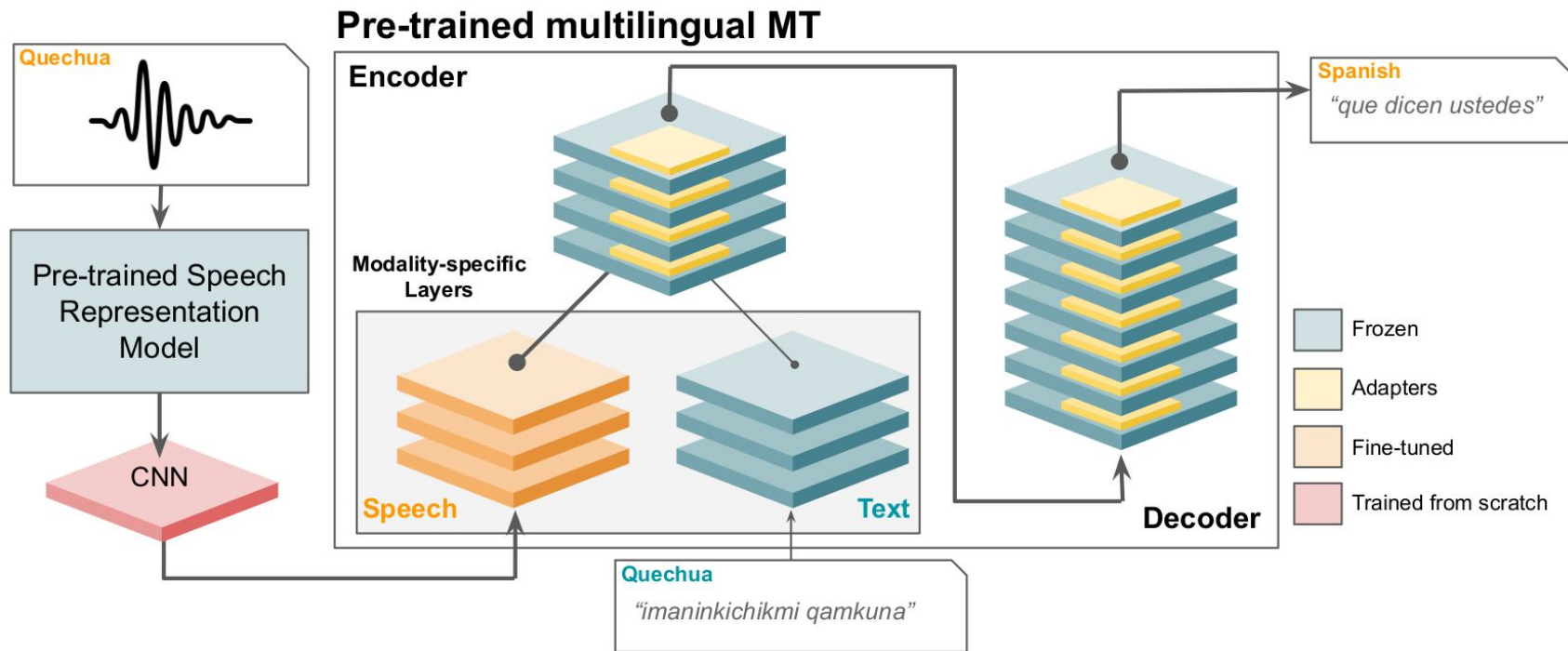


Figure: NLE architecture for ST and NMT

Training

- **SSL for speech:** Tamasheq-dedicated representation models and general-purpose multilingual models (XLSR-53; XLS-R)
 - Some degree of try-and-error needed to decide on the extraction layer
- **Pre-trained NMT:** NLLB 1-3B > mBART-50

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- **Speech data:**
 - **Low-resource ST:**
 - **17h** Tamasheq-French
 - **2h** Quechua-Spanish
 - **Low-resource ASR:**
 - **51h** Quechua

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 - **Low-resource ST:**
 - **17h** Tamasheq-French
 - **2h** Quechua-Spanish
 - **Low-resource ASR:**
 - **51h** Quechua
 - **High-resource ST:**
 - **187h** mTEDx (fr-en; es-fr; es-en; fr-es)
 - **High-resource ASR:**
 - **432h** mTEDx (fr; es)
 - **207h** TED-LIUM (en)

Results



SOTA Results

- **Our multilingual model reaches first place for both languages**
- For Tamasheq we are able to improve scores by using a ***dedicated-SSL***

SOTA evolution for Tamasheq-French IWSLT 2022 test set:

- Tamasheq-French IWSLT 2022 winner: **5.7 BLEU** [1]
- XLSR-SAMU post-evaluation submission: **13 BLEU** [2]
- NLE @ IWSLT 2023: **20.7 BLEU** [3] (IWSLT 2023 test set: 23.5 BLEU)

SOTA Results

- **Our multilingual model reaches first place for both languages**
- For Tamasheq we are able to improve scores by using a ***dedicated-SSL***
- **For Quechua we achieve 17 BLEU using only 2h of aligned translations**

One model for both high and low-resource

- On high-resource settings we are competitive against the winner of the IWSLT 2021 offline track

Them: Domain zero-shot language pairs

Model	Total params	Trained params	Training directions				Zero-shot directions		
			Es-En	Fr-En	Fr-Es	Pt-En	Pt-Es	It-En	It-Es
FAIR at IWSLT 2021 (Tang et al., 2021)	700M		40.4	36.4	34.4	29.0	34.4	28.4	34.6
	3×700M (ensemble)		42.2	38.7	36.5	31.0	38.2	29.4	37.3
XLS-R + NLLB 1.3B	317M + 1.38B	70M	43.7	39.4	38.0	31.5	35.9	28.9	35.0
XLS-R + NLLB 3.3B	317M + 3.36B	115M	44.0	39.9	38.3	33.1	38.1	29.3	36.9
XLS-R + NLLB 1.3B, ASR + MT cascade			41.8	35.6	34.4	29.7	35.8	29.3	35.2

Table: mTEDx performance (BLEU) at IWSLT 2021 offline track

Us: REAL zero-shot language pairs

One model for both high and low-resource

- 6x less trainable parameters

Them: Train everything from scratch



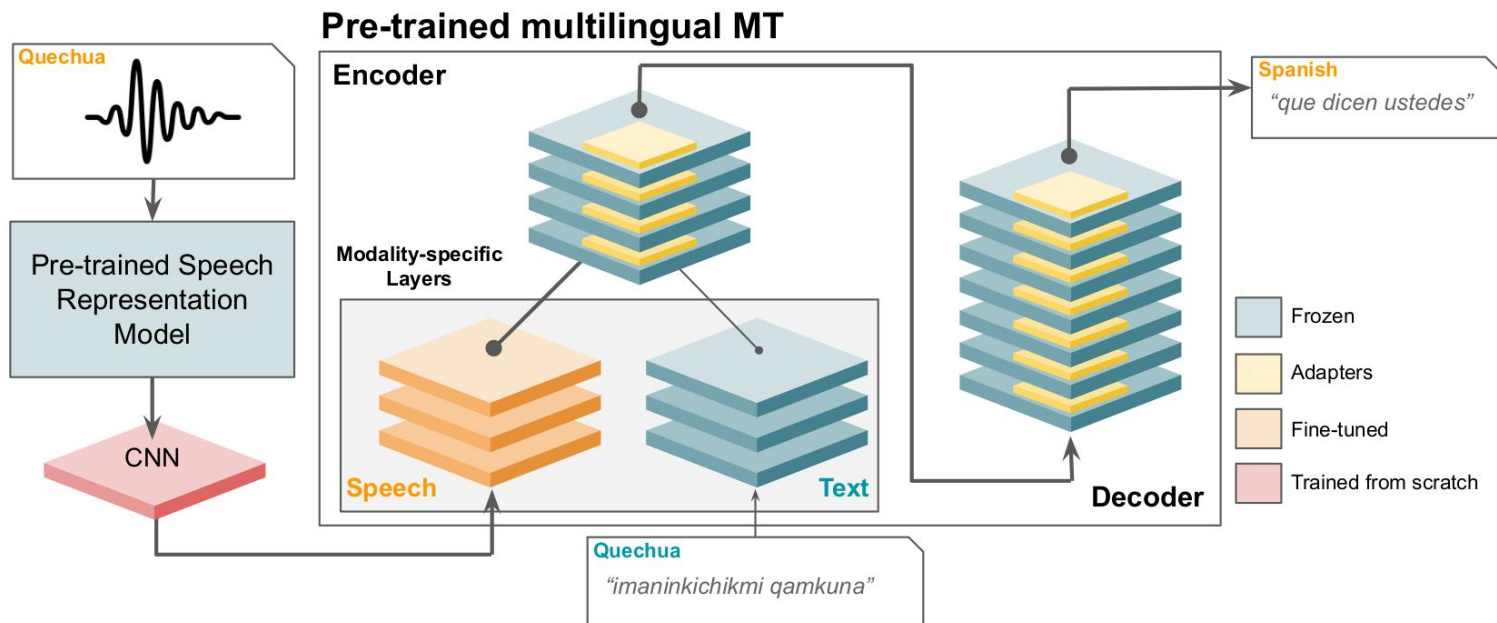
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Us: Train only interface and adapters

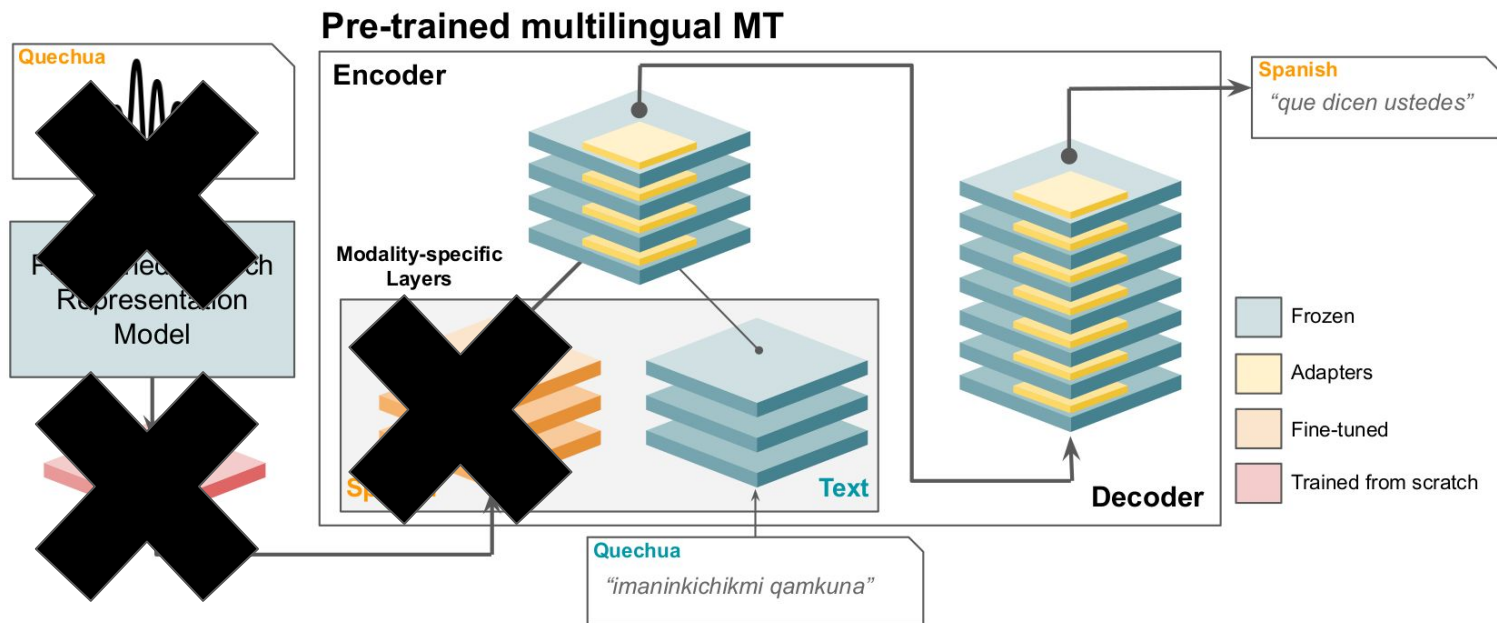
Some interesting characteristics: cross-modality domain transfer

- Domain adaptation for text with speech data



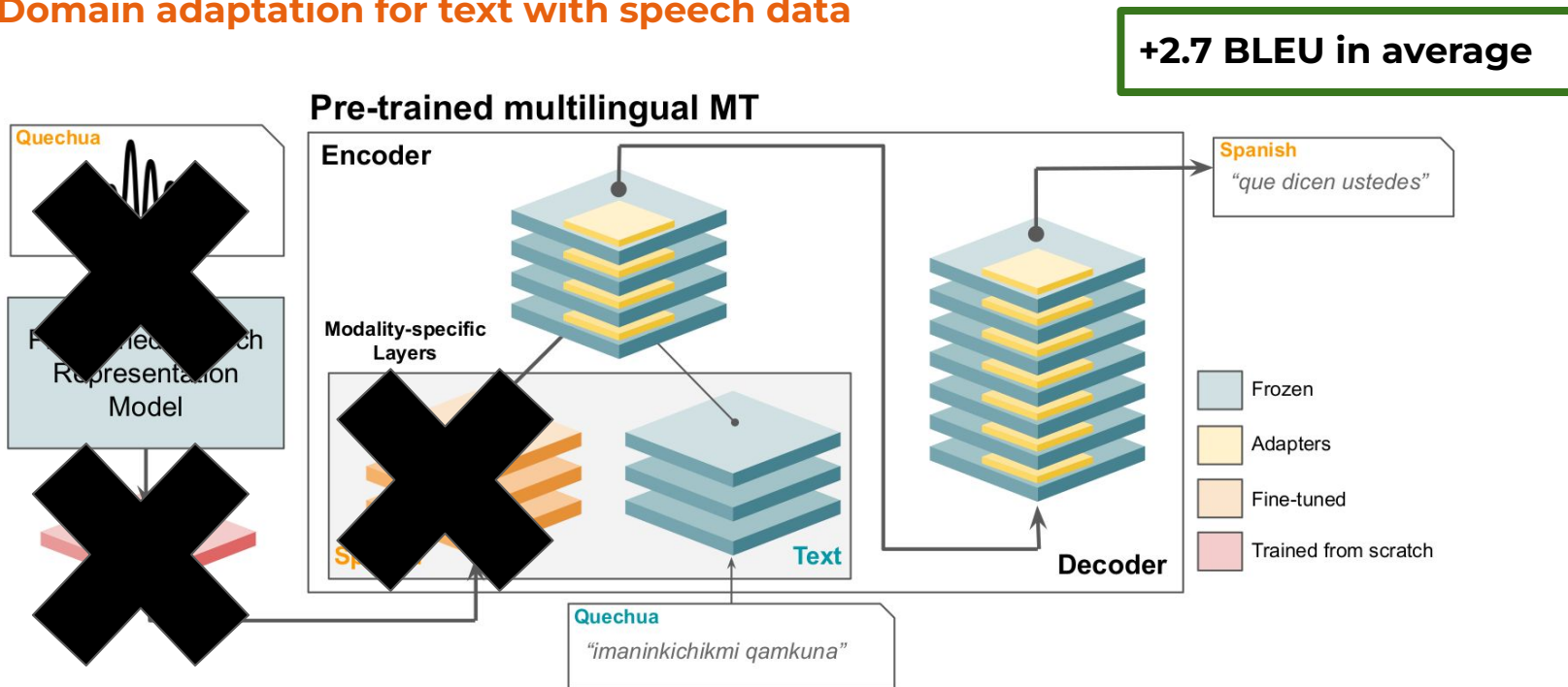
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- Domain adaptation for text with speech data



In summary

- We build a **multimodal** and **multilingual** model for both speech and text translation
- We adopt a **parameter-efficient** adapting strategy leveraging existing pre-trained models
- We are able to **achieve SOTA in low-resource ST (Tamasheq and Quechua)** and competitive results in high-resource settings with a flexible multilingual model



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Some interesting characteristics: incremental learning

2-step training with Tamasheq data only	Model	New params	Taq-Fr	These would still rank first in the IWSLT 2023 competition
	Joint training	0	21.06	
	Adapters 64 (all)	6.4M	17.60	
	Adapters 256 (all)	15.9M	18.18	
	Adapters 256 (bottom)	1.6M	19.24	
	Conv + Adapters 256 (bottom)	2.5M	19.13	

Table: Joint training (top) versus 2-step incremental learning for Tamasheq